# Computer Design.

## Von Neumann Architecture

1. Data & instr. stored in → single read-write memory.
2. Memory Contents → addressable by location.
3. Sequential execution.

PC → Program Counter
IR → Instruction Register
MAR → Memory address register
MBR → " Buffer "
I/O AR → Address Register
" BR → Buffer "

## Computer Function

Execute program → Using Processor

↓

2 steps

Fetch instr.

Decode + execute "

Instruction cycle – a complete processing required for a single instruction – consisting of fetch cycle and execute cycle – halts if machine is turned off, unrecoverable error,

HALT instruction

program counter → holds address of the next instr. to be fetched.

Instruction Register (IR) – Store the fetched instruction – specify the action (opcode and address)